

Shubham Mandora

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SUMMARY

Software engineer building production backend systems and data infrastructure in Python and Go. Founding-engineer experience designing ETL pipelines, pub/sub and REST services, and relational and NoSQL data models (PostgreSQL, MongoDB, DynamoDB) on AWS. Hands-on with observability (Prometheus, Grafana, OpenTelemetry), containerized deployment (Docker, Kubernetes, CI/CD), and RAG and agentic-AI systems using LangChain and the Claude API.

PROFESSIONAL EXPERIENCE

Vizzy Labs, Inc., *Founding Engineer - Backend and Data* 08/2025 – Present | Brooklyn, NY

- Built and operated production backend data infrastructure on AWS in Python, ingesting high-volume event data from internal and external sources with automated data-quality and integrity checks.
- Built observability and alerting for the platform with Prometheus, Grafana, and automated alerts, improving pipeline performance and catching issues before they reached consumers.
- Designed and optimized relational (PostgreSQL) and NoSQL (MongoDB, DynamoDB) data models and storage for downstream analytical and operational services at scale.
- Worked across product and operations engineering on reliability, testing frameworks, and shared documentation, using AI coding tools (Claude Code, Cursor) to accelerate delivery.

New York Institute of Technology, *Research Assistant* 01/2025 – 08/2025 | New York City, USA

- Built production Python data pipelines with Airflow-style orchestration over multi-source datasets, with automated cleaning, schema standardization, and quality validation.
- Integrated and tested scikit-learn models (Random Forest, SVM, MLP) for data classification and deduplication, with hyperparameter tuning and cross-validation. (*Trulioo removed*)
- Built RAG-based semantic search and retrieval pipelines using LangChain and pgvector with embedding-based similarity search.
- Designed PostgreSQL- and MongoDB-backed data models with structured validation and ETL workflows, plus reproducible experiment frameworks and documentation.
- Developed multi-channel real-time ingestion and visualization pipelines processing high-frequency sensor streams into structured outputs.

IDZ Digital Pvt. Ltd., *Software Developer Intern* 02/2022 – 04/2022 | Mumbai, India

- Developed consumer mobile games across the full SDLC in Java, Kotlin, C#, and Unity, shipping to the App Store and Play Store.
- Implemented backend game logic and automated validation scripts for gameplay consistency across iterative releases.
- Designed and ran regression and performance test plans to catch defects early, contributing to team documentation.

Clear Exam, *Python Developer Intern* 05/2021 – 07/2021 | Nashik, India

- Built a high-throughput data processing pipeline in Java and Python with automated classification workflows and structured testing.
- Designed secure API-driven backend services with authentication, input validation, and data-integrity checks.

Divspace, LLP, *Software Developer Intern* 07/2020 – 10/2020 | Nasik, India

- Developed Python and JavaScript automation scripts for property-listing data processing and API integration on a real estate platform.
- Built CI/CD-style deployment workflows with automated data-integrity checks for consistent releases.

EDUCATION

New York Institute of Technology, *Master's of Science in Computer Science* 09/2023 – 05/2025

Sandip University, *Bachelor of Technology in Computer Science* 07/2019 – 05/2022 | Nasik, India

PROJECTS

Online Proctoring Using AI

- Built a real-time online-proctoring system with a React/TypeScript frontend running MediaPipe face landmarking at ~15 FPS and a FastAPI backend running YOLOv8 (object and multi-person detection), InsightFace ArcFace (government-ID identity recheck), and Silero VAD, streaming deduplicated events over WebSocket at ~5–20 KB/s.
- Implemented gaze tracking with per-user 5-station calibration plus head-pose, talking, multi-monitor, tab-switch, and object (phone/book/laptop) detection feeding a live per-session suspicion score and a proctor dashboard with evidence clips and Whisper transcription of flagged speech.
- Hardened the service with Fernet-encrypted ID storage at rest, JWT-secret boot enforcement, and per-IP login rate limiting; added exam management with MCQ auto-grading.

AI Code Review Agent

- Built an autonomous GitHub code-review agent as a single Docker container (FastAPI webhook receiver, background worker, LangGraph state machine, server-rendered dashboard) with Postgres/pgvector as the only external dependency.
- Implemented the review flow: HMAC-verified pull_request webhooks enqueued with a sub-second 202, then a LangGraph graph that fetches the diff, runs a preflight cost estimate, retrieves repo conventions from pgvector (Voyage AI embeddings), and runs four parallel Claude check nodes (correctness, security, style, test coverage).

- Deduplicated findings into one GitHub review per PR head SHA, achieving a p95 review target under 30 seconds on a 200-line diff at under \$0.50 per review, gated by an evaluation suite of hand-labeled PRs (80%+ findings match); configurable per-repo via `.codereview.yml`.

Real-time Pub/Sub Chat Service (Go)

- Built a horizontally scalable real-time chat backend in Go: stateless WebSocket gateway replicas fan messages out via Redis pub/sub, a separate worker persists to Postgres, and the gateway serves history, presence, typing, auth, and rate limiting (30 msg/min Redis token bucket).
- Added graceful shutdown, readiness gating (`/readyz` returns 503 when Redis or Postgres is unreachable), paginated history with replay since last-seen ID, and public/private rooms with invite tokens.
- Instrumented every service with Prometheus metrics (active connections, messages/sec, fan-out latency histogram) and OpenTelemetry tracing to Jaeger; tested with table-driven unit tests and Redis/Postgres integration tests via `testcontainers-go`.
- Shipped Kubernetes manifests with an HPA autoscaling on active connections per pod (min 2 / max 20), and a k6 load test ramping to 10k concurrent connections against a p99 fan-out latency target under 50ms.

Sustainable City Management

- Built a real-time data platform for Dublin city infrastructure, ingesting live DublinBikes, Sonitus noise-sensor, and OpenWeather feeds via Celery workers and serving them over REST + Server-Sent Events to three live React (Vite + Leaflet + Recharts) dashboards.
- Backed it with Django 5 + DRF + PostgreSQL 16 + Redis 7, with health/readiness probes, structured JSON logging, and Grafana + Prometheus observability.
- Set up local Kubernetes staging on kind behind nginx-ingress mirroring production topology, with GitHub Actions CI/CD and a test suite spanning pytest integration, Playwright e2e, and k6 load tests.

SLO Reference Observability Stack

- Built an end-to-end SLO observability slice: a Go `orders-api` with `env-var` chaos fault injection (error rate and latency), scraped by Prometheus, with SLI, error budget, multi-window burn rate, and p95 computed as Prometheus recording rules.
- Exposed SLO state through a Go BFF (`GET /api/slo`) rendered live from one source of truth in two surfaces: a custom React (Vite + Tailwind + Recharts) SLO Control Center and a Grafonnet-generated Grafana dashboard with zero manual UI editing.
- Defined the `orders-api` SLO as 99.9% availability over 28 days at 200ms p95, drove it with deterministic k6 load, and unit-tested the Prometheus recording rules with `promtool`.

SKILLS

Programming Languages — Python | Go | TypeScript | JavaScript | Java | Kotlin | C# | SQL | Shell

Backend & APIs — FastAPI | Django | Django REST Framework | Celery | REST APIs | WebSockets | Server-Sent Events

Frontend — React | Vite | Tailwind CSS | Recharts | Leaflet | HTML/CSS

Data and Storage — PostgreSQL | Redis | MongoDB | DynamoDB | pgvector | Airflow | ETL pipelines | pandas

Cloud, DevOps & Infrastructure — AWS (EC2, S3, SageMaker, Lambda) | GCP | Azure | Docker | Docker Compose | Kubernetes (HPA) | GitHub Actions | CI/CD

Observability & Testing — Prometheus | Grafana | Grafonnet | OpenTelemetry | Tempo | Loki | Jaeger | k6 | pytest | Playwright | `testcontainers-go` | `promtool`

AI and ML — LangChain | LangGraph | RAG pipelines | Agentic workflows | Claude API | Vector embeddings | PyTorch | scikit-learn | Random Forest | SVM | XGBoost | YOLOv8 | MediaPipe | Whisper

Tools — Git | GitHub | Linux (Ubuntu) | Jira | Confluence | Claude Code | Cursor

CERTIFICATES

Google Advanced Data Analytics (Coursera) | **Python Essentials Course badge** (Cisco) | **AWS certified Cloud Practitioner** (AWS)